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Practical : Apply filtering, sorting, aggregation, and GroupBy operations on real-world datasets.

step 1 : import library

```
# data set

import pandas as pd
import numpy as np
```

create a dataset and dataframe

```
data = {
    "Emp_id":[101,102,103,104,105,106,107,108],
    "Name":["Rahul","Priya","Amit","Riya","karan","Neha","Vivek","Pooja"],
    "Department":["IT","HR","IT","Finance","HR","Finance","IT","HR"],
    "City":["Surat","Navsari","Surat","Valsad","Surat","Navsari","Valsad","Surat"],
    "Salary":[45000,38000,52000,48000,40000,51000,60000,42000],
    "Experience":[2,1,5,4,2,6,8,3]
}
```

```
df=pd.DataFrame(data)
df
```

	Emp_id	Name	Department	City	Salary	Experience
0	101	Rahul	IT	Surat	45000	2
1	102	Priya	HR	Navsari	38000	1
2	103	Amit	IT	Surat	52000	5
3	104	Riya	Finance	Valsad	48000	4
4	105	karan	HR	Surat	40000	2
5	106	Neha	Finance	Navsari	51000	6
6	107	Vivek	IT	Valsad	60000	8
7	108	Pooja	HR	Surat	42000	3

Next steps: Generate code with df | New interactive sheet

step 2: filter the data

step 2.1: display the employee from IT department

```
# Step 2 : Filter Data
df[df["Department"]=="IT"]
```

	Emp_id	Name	Department	City	Salary	Experience
0	101	Rahul	IT	Surat	45000	2
2	103	Amit	IT	Surat	52000	5
6	107	Vivek	IT	Valsad	60000	8

Step 2.2:display employees salary over 50000

```
# dipaly salary over 50000
df[df["Salary"]>50000]
```

	Emp_id	Name	Department	City	Salary	Experience
2	103	Amit	IT	Surat	52000	5
5	106	Neha	Finance	Navsari	51000	6
6	107	Vivek	IT	Valsad	60000	8

Step 2.3 display employee from surat

```
#display employess from surat city
df[df["City"]=="Surat"]
```

	Emp_id	Name	Department	City	Salary	Experience
0	101	Rahul	IT	Surat	45000	2
2	103	Amit	IT	Surat	52000	5
4	105	karan	HR	Surat	40000	2
7	108	Pooja	HR	Surat	42000	3

Step 2.4: display employee from IT deparment haviing salary more than 50000

```
#display employee from IT deparment haviing salary more than 50000
df[(df["Department"]=="IT")&(df["Salary"]>50000)]
```

	Emp_id	Name	Department	City	Salary	Experience
2	103	Amit	IT	Surat	52000	5
6	107	Vivek	IT	Valsad	60000	8

step 3: sorting data

Step 3.1:sorting salary in ascending order

```
#ascending order
df.sort_values(by="Salary")
```

	Emp_id	Name	Department	City	Salary	Experience
1	102	Priya	HR	Navsari	38000	1
4	105	karan	HR	Surat	40000	2
7	108	Pooja	HR	Surat	42000	3
0	101	Rahul	IT	Surat	45000	2
3	104	Riya	Finance	Valsad	48000	4
5	106	Neha	Finance	Navsari	51000	6
2	103	Amit	IT	Surat	52000	5
6	107	Vivek	IT	Valsad	60000	8

Step 3.2:sorting salary in descending order

```
#descending order
df.sort_values(by="Salary", ascending=False)
```

	Emp_id	Name	Department	City	Salary	Experience
6	107	Vivek	IT	Valsad	60000	8
2	103	Amit	IT	Surat	52000	5
5	106	Neha	Finance	Navsari	51000	6
3	104	Riya	Finance	Valsad	48000	4
0	101	Rahul	IT	Surat	45000	2
7	108	Pooja	HR	Surat	42000	3
4	105	karan	HR	Surat	40000	2
1	102	Priya	HR	Navsari	38000	1

Step 4:perfrom aggregation

Step 4.1: Calculate total salary of employee

```
#employee salary total
df["Salary"].sum()
```

```
np.int64(376000)
```

Step 4.2: calculate the avg salary

```
#avg salary
df["Salary"].mean()
```

```
np.float64(47000.0)
```

Step 4.3: find the highest salary

```
#highest salary
df["Salary"].max()
```

```
60000
```

Step 4.4: find the lowest salary

```
# lowest salary
df["Salary"].min()
```

```
38000
```

Step 5 : perfrom Group by operations

step 5.1 group employees based on department& calculate avg salary of each department

```
# Group by employee and calculate avg salary based on department
df.groupby("Department")["Salary"].mean()
```

Department	Salary
Finance	49500.000000
HR	40000.000000
IT	52333.333333

dtype: float64

5.2 calculate the total slary paid by each deparment

```
#Total salary
df.groupby("Department") ["Salary"].sum()
```

Department	Salary
Finance	99000
HR	120000

```
# display the highest salary in each deparment
df.groupby("Department")["Salary"].max()
```

Department	Salary
Finance	51000
HR	42000
IT	60000

dtype: int64

```
#calculate the no of employee in each department
df.groupby("Department")["Salary"].count()
```

Department	Salary
Finance	2
HR	3
IT	3

dtype: int64

Step 6: multiple Aggregation function

```
#using multiple aggregation
df.groupby("Department")["Salary"].agg(["mean","sum","max","min","count"])
```

Department	mean	sum	max	min	count
Finance	49500.000000	99000	51000	48000	2
HR	40000.000000	120000	42000	38000	3
IT	52333.333333	157000	60000	45000	3

Step 7:Save your result

```
# Saving result
df.to_csv("employee_data.csv",index=False)
print("Data_set saved successfully")
```

```
Data_set saved successfully
```

Step 8: download csv file

```
# Downloading csv file
from google.colab import files
files.download("employee_data.csv")
```